

AssureX Claim Engine

Google Teachable Machine Report

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Live application	https://assurex.pythonanywhere.com
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Abstract

AssureX Claim Engine evaluates warranty claims for consumer electronics, home appliances and industrial tools. A customer or service center registers a product from its receipt (read by OCR), files a claim with evidence, and the application pre-processes the claim into model features. A calibrated Python classification model (HistGradientBoosting, 89.3% test accuracy on 225 unseen claims) scores the structured record; the same claim is drawn as a Claim Summary Card and classified by a Google Teachable Machine model (93.8% test accuracy). The two predictions and their confidences are compared (Strong Match, Acceptable Match, Weak Match, Model Disagreement, Uncertain Result), the category's warranty policy is executed from JSON configuration, contradictions, missing documents and duplicates are detected, and a decision table produces Likely Valid, Likely Invalid or Manual Review Required. Anything uncertain goes to a human reviewer, and every step is recorded in an audit trail. The application has four roles (customer, service-center staff, claim reviewer, administrator), 295 automated tests and 34 browser tests.

Evidence for the Google Teachable Machine model (SRS §1.10 item 5): classes and samples, training configuration and observations, incorrectly classified samples, retraining, the exported model and label file, and screenshots of training, testing and evaluation.

1. Google Teachable Machine evidence

Status: trained and installed. Model [gtm-514702678b8c](#), trained in the browser on the 2,100 v3 training cards: **93.8% test accuracy** (macro F1 93.8%), 97.3% validation accuracy, agreeing with the Python model on 92.9% of test claims. A first model trained on the v2 cards scored 54.2% and was replaced (see *Card design iterations*). The earlier [gtm_classifier.joblib](#) / [weights.bin](#) / [metadata.json](#) files were a scikit-learn stand-in, not a Teachable Machine export, and were removed.

Project evidence (in place of a project link)

The SRS accepts "Google Teachable Machine project link **or** permitted project evidence". The model was trained in the browser and exported directly; the browser project was not saved to Google Drive, so there is no shareable project link. The following evidence is submitted instead, and every item can be checked against the others:

Evidence	Where
Exported Teachable Machine model (TensorFlow Lite, float)	model/teachable_machine/model_unquant.tflite ; its SHA-256 begins 514702678b8c , which is the model version gtm-514702678b8c shown in the application and stored with every prediction
Label file (the three classes, in Teachable Machine's order)	model/teachable_machine/labels.txt
Screenshots of the three classes, training and in-browser testing	screenshots/23_gtm_v3_training.png (the three classes, 700 samples each, model trained), screenshots/24_gtm_v3_preview_test.png (preview test), screenshots/25_gtm_v3_evaluation.png (evaluation); first run: screenshots/20_gtm_training.png , screenshots/22_gtm_evaluation.png
Class samples (Valid Claim, Invalid Claim, Manual Review)	the training cards in data/summary_cards/train/{valid,invalid,manual_review}/ (700 per class) and the validation and test cards in data/summary_cards/val/ and data/summary_cards/test/
Training configuration and retraining details	<i>Training record</i> below (epochs 50, batch 16, learning rate 0.001; run 1 on v2 cards, run 2 on v3 cards)
Evaluation on unseen cards	model/teachable_machine/evaluation.json (225 validation and 225 test cards: 97.3% and 93.8%)
Incorrectly classified samples	reports/gtm_misclassified_test.csv , reports/gtm_misclassified_val.csv
Comparison with the Python model	reports/model_comparison_report.md and .csv (225 test claims)

Reproduce the figures: [python notebooks/evaluate_gtm.py](#) re-scores the installed model on the 225 validation and 225 test cards, or sign in as the administrator and use **Admin > Models > Run evaluation**. The test suite also re-scores the model on every run and fails if its accuracy differs from the recorded 93.8% ([tests/test_ml_integrity.py](#)).

Claim Summary Cards

Item	Value
Renderer	src/core/card_v2.py (same code for training cards and live claims)
Size	600 × 600 px (square, so Teachable Machine's centre crop keeps every element)
Fonts	DejaVu Sans shipped in static/fonts/ (identical rendering on Windows / Linux / servers)

Item	Value
Content (v3)	Claim facts compared with the category's policy limits, as a fixed 3×3 grid of large tiles: coverage (within / grace / ended), reporting time (in time / late), damage cause (covered / excluded, with diagnostic confidence), receipt, supporting evidence (3 of 3 ... 0 of 3), serial (matches / differs / unverified), dates (consistent / conflict), invoice (unique / reused), repairs (count / repeat / unauthorised). Each state has its own colour and glyph (✓ ! × ↔); a colour band shows the category. Limits come from policies/*.json , the same source as the rule engine
Never on a card	Python prediction, any confidence score, final decision, class label
Training images	2 variations per training claim (colour jitter, date format, ±1.5° rotation, blur, JPEG quality) → 2,100 images: 700 per class in data/summary_cards/train/{valid,invalid,manual_review}/
Validation / test	one canonical card per claim (<claim_id>_v0.jpg) in data/summary_cards/{val,test}/ — never uploaded for training
Mapping	data/claim_id_to_card_mapping.csv (claim ID, split, class, variation, filename, path)

Samples: Admin > Models shows one training card per class; any card can be opened from a claim page.

Card design iterations (measured)

Card	How measured	Validation	Test	Notes
v2 (grey bars, small tiles and pips)	Real Teachable Machine export gtm-ec95b4aa49b5 , evaluated in the app	52.9%	54.2% (macro F1 45.1%)	agreed with the Python model on 55.1% of test claims; below the SRS 85% target
v2	notebooks/tm_replica_check.py , 2 seeds	63.1–74.7%	58.2–65.8%	the replica reproduces the failure, so the card, not the training run, was the problem
v3 (policy-relative colour tiles)	notebooks/tm_replica_check.py , 3 seeds	95.1–97.3%	92.9–94.2% (mean 93.6%)	pre-check before the browser run
v3	Real Teachable Machine export gtm-514702678b8c , evaluated in the app	97.3%	93.8% (macro F1 93.8%)	agrees with the Python model on 92.9% of test claims; ceiling ≈ 96% (4% deliberate label noise)

Why v2 failed: Teachable Machine does not fine-tune the image network; it trains a small head on frozen ImageNet (MobileNet) features. Thin grey bars and 20-pixel pips at 224×224 produce almost identical features for all three classes. v3 makes every fact a large region whose colour and glyph change with the fact's value relative to the policy limit, which those frozen features separate easily. The card still shows facts only — no class, decision, rule outcome or model confidence.

The replica estimate (92.9–94.2%) matched the real v3 export (93.8%), so the offline check is a useful way to test a card design before spending a browser training run.

Training procedure

1. Admin > Models > *Download training cards* (zip with three class folders), or use the folders above.
2. [teachablemachine.withgoogle.com](#) → *Image Project* → *Standard image model*.
3. Rename the classes exactly **Valid Claim**, **Invalid Claim**, **Manual Review**; upload 700 images each.
4. *Advanced*: start with epochs 50, batch size 16, learning rate 0.001. Train.
5. *Export Model* → *Tensorflow Lite* → *Floating point* → download; unzip.

6. Admin > Models > upload `model_unquant.tflite` and `labels.txt` (checked: TFLite header, exactly the three labels, the model loads and runs; the previous model is archived under `model/teachable_machine/versions/`).
7. Run *evaluation* (or `python notebooks/evaluate_gtm.py`) and `python reports/generate_comparison_report.py`.

Confusion matrix (test, 225 cards; rows = actual, columns = predicted)

	Valid Claim	Invalid Claim	Manual Review
Valid Claim	73	1	1
Invalid Claim	5	68	2
Manual Review	4	1	70

Most errors predict *Valid Claim* for claims labelled Invalid or Manual Review (9 of 14). 12 of the 14 errors are claims whose label was deliberately flipped, where the model's answer is the policy's answer (see *Training observations*); in the application no claim is marked *Likely Valid* unless the rule engine and the Python model agree as well.

Training observations

- **Run 1 (v2 cards).** Training finished normally (epochs 50, batch 16, learning rate 0.001), but on our 225 test cards the model reached only 54.2% and agreed with the Python model on 55.1% of claims: a card that looks clear to a person is not necessarily clear to frozen ImageNet features at 224 × 224 pixels.
- **Diagnosis.** An offline replica of the trainer (`notebooks/tm_replica_check.py`) reproduced the failure (58–66%), so the problem was the input representation, not the training run or its settings.
- **Run 2 (v3 cards).** Same settings. Validation 97.3%, test 93.8%; the preview in Teachable Machine gave *Invalid Claim* 99% for an unseen card with a date conflict, late reporting and an excluded cause (`screenshots/24_gtm_v3_preview_test.png`).
- **Where it still errs.** 12 of its 14 test errors are claims whose label was deliberately flipped to simulate reviewer disagreement (the test split has 12 such claims): on every one of them the image model gave the answer the warranty policy gives. Its only 2 real errors are a grace-period claim called *Invalid* (should be reviewed) and a late-reporting claim called *Manual Review* (should be invalid). Measured against the policy instead of the noisy labels it agrees on 223 of 225 test claims (99.1%). The misclassified cards are listed with their scenario in `reports/gtm_misclassified_test.csv`.
- **Browser behaviour.** With 2,100 images Chrome showed "Page unresponsive" during "Preparing training data"; choosing *Wait* and keeping the tab in front let training finish (about 20–25 minutes).
- **Deployment.** TensorFlow 2.20 no longer ships `tf.lite.Interpreter`; the app uses `ai-edge-litert` (Windows and Linux). Models are loaded from bytes so a new export can replace a running one on Windows.

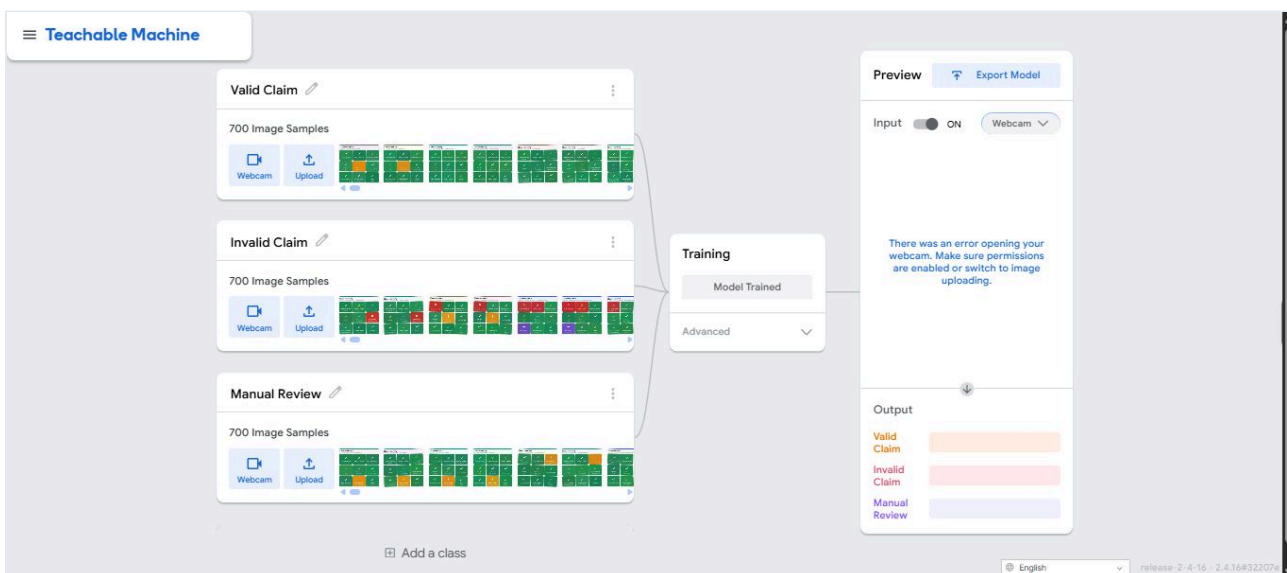
Runtime

`src/core/gtm_classifier_v2.py` — TensorFlow Lite interpreter (`ai-edge-litert`), preprocessing identical to the Teachable Machine export sample (ImageOps.fit to the model input, scaled to [-1, 1]; quantized exports are supported). The model version is `gtm-<SHA-256 prefix of the .tflite file>` and is stored with every prediction. If the file is missing or broken the prediction is marked *unavailable* and the claim goes to manual review — there is no fallback model.

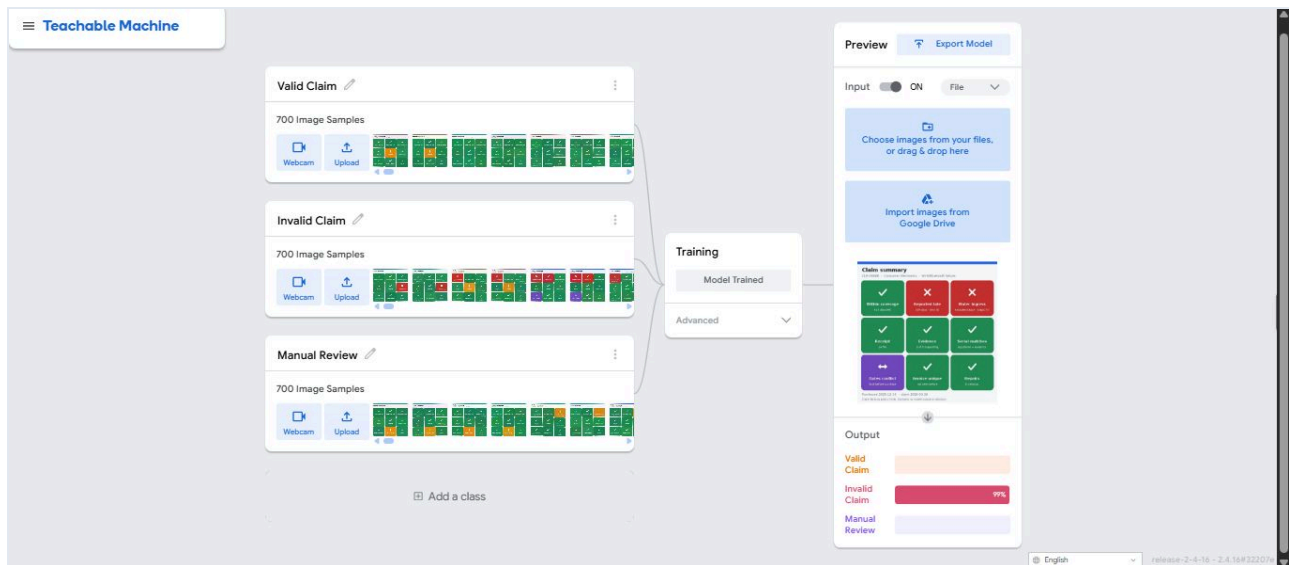
Training record

Item	Value
Model version	gtm-514702678b8c (model/teachable_machine/model_unquant.tflite, float, input 224×224)
Classes	Valid Claim · Invalid Claim · Manual Review (labels.txt)
Images per class	700 / 700 / 700 v3 training cards (two variations of each of the 1,050 training claims)
Training configuration	Standard image model; epochs 50, batch size 16, learning rate 0.001
Validation accuracy (225 cards)	97.3%
Test accuracy (225 cards) · macro F1	93.8% · 93.8%
Agreement with the Python model (test)	92.9% of claims
Macro precision / recall (test)	94.0% / 93.8%
Per class (test): precision / recall	Valid 89.0% / 97.3% · Invalid 97.1% / 90.7% · Manual Review 95.9% / 93.3%
Full evaluation	model/teachable_machine/evaluation.json; comparison with the Python model in reports/model_comparison_report.md (application decision accuracy 92.0%)
Incorrectly classified samples	reports/gtm_misclassified_test.csv (written by the evaluation)
Retraining details	Run 1 on v2 cards → 54.2% test (archived as gtm-ec95b4aa49b5). Cause found with notebooks/tm_replica_check.py; card redesigned (v3); run 2 on v3 cards → 93.8%
Screenshots	Run 1: screenshots/20_gtm_training.png, 22_gtm_evaluation.png. Run 2: 23_gtm_v3_training.png, 24_gtm_v3_preview_test.png, 25_gtm_v3_evaluation.png

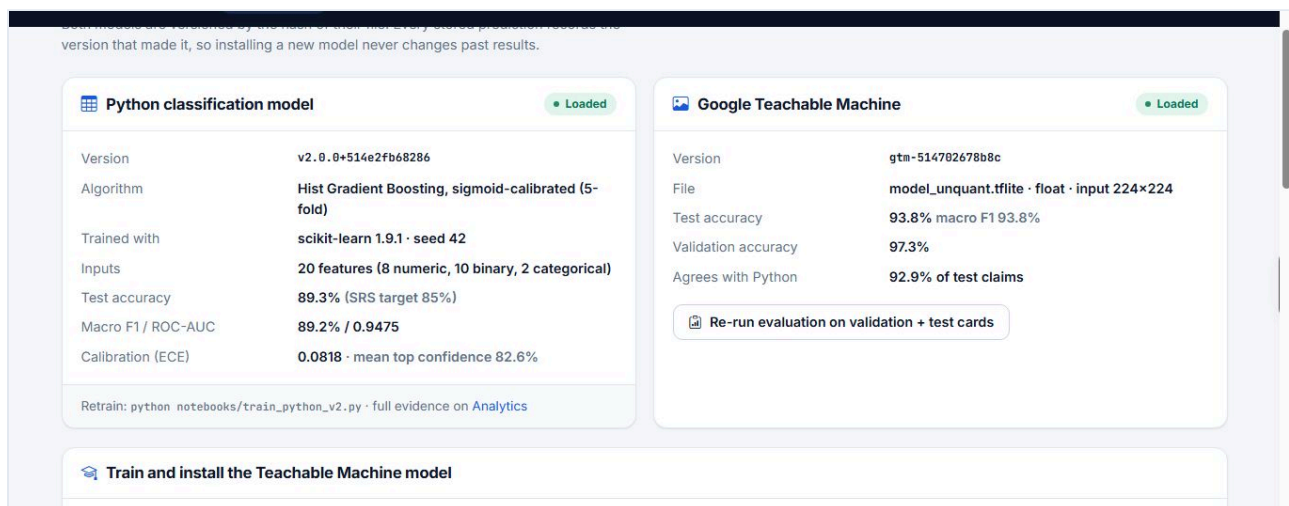
Teachable Machine screenshots



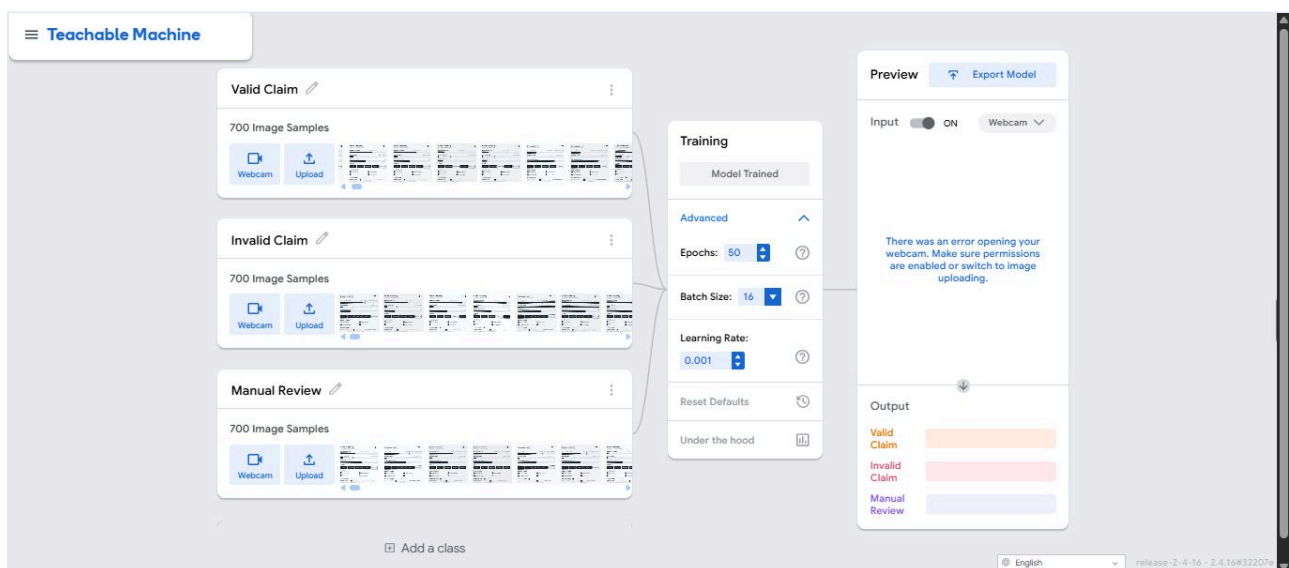
Run 2 (v3 cards): the three classes, 700 images each, model trained



Run 2: preview test on an unseen card



Run 2: evaluation of the installed model in the application



Run 1 (v2 cards): training

CLAIM ENGINE

Both models are versioned by the hash of their file. Every stored prediction records the version that made it, so installing a new model never changes past results.

Python classification model

Loaded

Version

v2.0.0+514e2fb68286

Algorithm

Hist Gradient Boosting, sigmoid-calibrated (5-fold)

Trained with

scikit-learn 1.9.1 · seed 42

Inputs

20 features (8 numeric, 10 binary, 2 categorical)

Test accuracy

89.3% (SRS target 85%)

Macro F1 / ROC-AUC

89.2% / 0.9475

Calibration (ECE)

0.0818 · mean top confidence 82.6%

Retrain: python notebooks/train_python_v2.py · full evidence on Analytics

Google Teachable Machine

Loaded

Version

gtm-ec95b4aa49b5

File

model_unquant.tflite · float · input 224×224

Test accuracy

54.2% macro F1 45.1%

Validation accuracy

52.9%

Agrees with Python

55.1% of test claims

Re-run evaluation on validation + test cards

Train and install the Teachable Machine model

Run 1: evaluation (54.2% test accuracy, replaced)

Class samples

Three training cards per class from [data/summary_cards/train/](#) (700 per class were uploaded to Teachable Machine).

Valid Claim

Claim summary

CLM-00004 · Consumer Electronics · Motherboard failure

✓

Within coverage

18 days left

✓

Reported in time

28 days · limit 30

✓

Manufacturing Defect

covered cause · diag 0.75

✓

Receipt

on file

✓

Evidence

3 of 3 supporting

✓

Serial matches

registered = evidence

✓

Dates consistent

purchase < fault < claim

✓

Invoice unique

not seen before

✓

Repairs

0 previous

Purchased 15/02/2024 · claim 28/01/2026

Claim facts vs policy limits. Contains no model output or decision.

CLM-00004_v1.jpg

Claim summary

CLM-00004 · Consumer Electronics · Motherboard failure

✓

Within coverage

18 days left

✓

Reported in time

28 days · limit 30

✓

Manufacturing Defect

covered cause · diag 0.75

✓

Receipt

on file

✓

Evidence

3 of 3 supporting

✓

Serial matches

registered = evidence

✓

Dates consistent

purchase < fault < claim

✓

Invoice unique

not seen before

✓

Repairs

0 previous

Purchased 15 Feb 2024 · claim 28 Jan 2026

Claim facts vs policy limits. Contains no model output or decision.

CLM-00004_v2.jpg

Claim summary

CLM-00005 · Consumer Electronics · Motherboard failure

✓

Within coverage

120 days left

✓

Reported in time

6 days · limit 30

✓

Manufacturing Defect

covered cause · diag 0.91

✓

Receipt

on file

✓

Evidence

3 of 3 supporting

✓

Serial matches

registered = evidence

✓

Dates consistent

purchase < fault < claim

✓

Invoice unique

not seen before

✓

Repairs

0 previous

Purchased 18/10/2025 · claim 20/06/2026

Claim facts vs policy limits. Contains no model output or decision.

CLM-00005_v1.jpg

Invalid Claim

Claim summary

CLM-00006 · Home Appliances · Thermostat failure

✗

Coverage ended

243 days ago (grace 14)

✓

Reported in time

4 days · limit 45

!

Accidental Drop

excluded cause · diag 0.19

✓

Receipt

on file

✓

Evidence

3 of 3 supporting

✓

Serial matches

registered = evidence

✓

Dates consistent

purchase < fault < claim

✓

Invoice unique

not seen before

✓

Repairs

1 previous

Purchased 26/06/2023 · claim 24/02/2026

Claim facts vs policy limits. Contains no model output or decision.

CLM-00006_v1.jpg

Claim summary

CLM-00006 · Home Appliances · Thermostat failure

✗

Coverage ended

243 days ago (grace 14)

✓

Reported in time

4 days · limit 45

!

Accidental Drop

excluded cause · diag 0.19

✓

Receipt

on file

✓

Evidence

3 of 3 supporting

✓

Serial matches

registered = evidence

✓

Dates consistent

purchase < fault < claim

✓

Invoice unique

not seen before

✓

Repairs

1 previous

Purchased 26 Jun 2023 · claim 24 Feb 2026

Claim facts vs policy limits. Contains no model output or decision.

CLM-00006_v2.jpg

Claim summary

CLM-00011 · Industrial Tools · Gearbox seizure

✓

Within coverage

15 days left

✓

Reported in time

23 days · limit 30

✓

Normal Wear and Tear

covered cause · diag 0.44

✓

Receipt

on file

✓

Evidence

3 of 3 supporting

✓

Serial matches

registered = evidence

✓

Dates consistent

purchase < fault < claim

✓

Invoice unique

not seen before

✗

Unauthorised repair

1 previous repair(s)

Purchased 17/01/2023 · claim 02/01/2026

Claim facts vs policy limits. Contains no model output or decision.

CLM-00011_v1.jpg

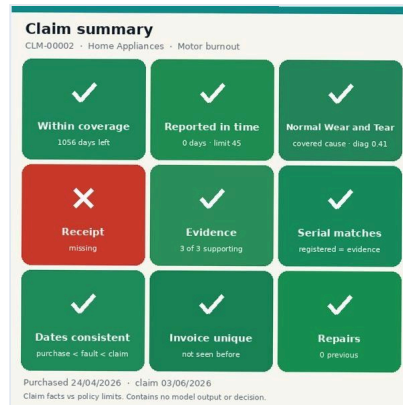
Manual Review



CLM-00001_v1.jpg



CLM-00001_v2.jpg



CLM-00002_v1.jpg